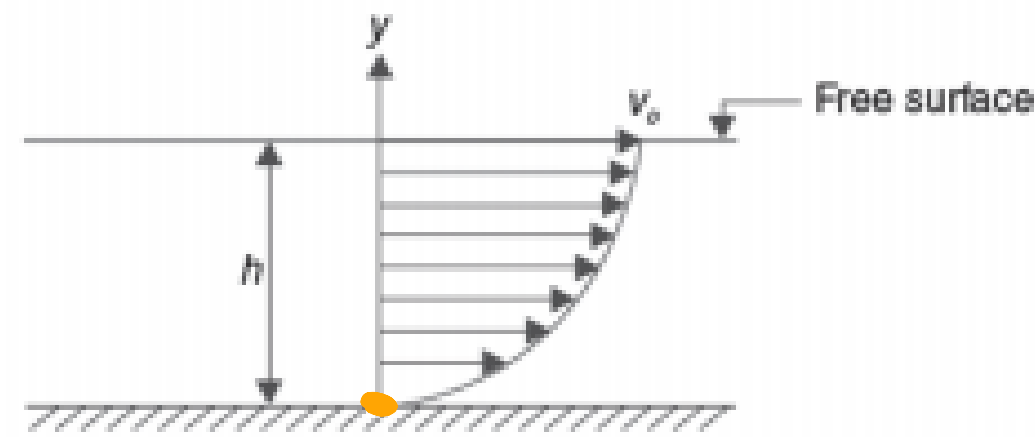


Q.2. A liquid is flowing through a horizontal channel. The speed of flow (v) depends on height (y) from the floor as $v = v_0 \left[2 \left(\frac{y}{h} \right) - \left(\frac{y}{h} \right)^2 \right]$. Where h is the height of liquid in the channel and v_0 is the speed of the top layer. Coefficient of viscosity is η . Calculate the shear stress that the liquid exerts on the floor.

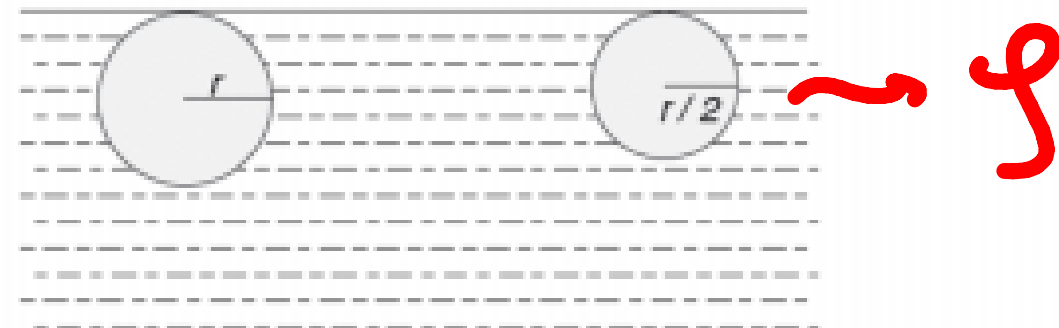


$$\frac{F_v}{A} = \frac{\eta A \frac{dv}{dy}}{A} = \eta \frac{dv}{dy}$$

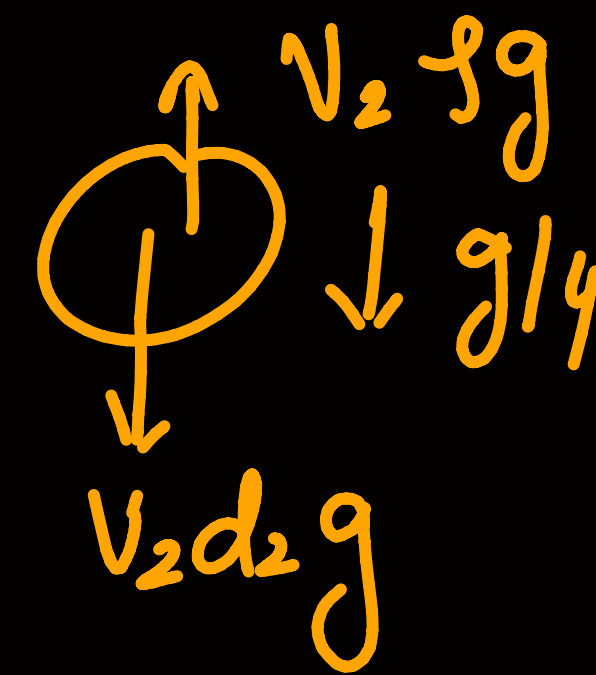
$$= \eta \int v_0 \left(\frac{2}{h} - \frac{1 \times 2y}{h^2} \right) dy$$

$$= \frac{\eta v_0 2}{h} = \underline{\text{shear stress.}}$$

Q.6. Two balls of radii r and $\frac{r}{2}$ are released inside a deep water tank. Their initial accelerations are found to be $\frac{g}{2}$ and $\frac{g}{4}$ respectively. Find the velocity of smaller ball relative to the larger ball, a long time after the two balls are released. Coefficient of viscosity is given to be η .



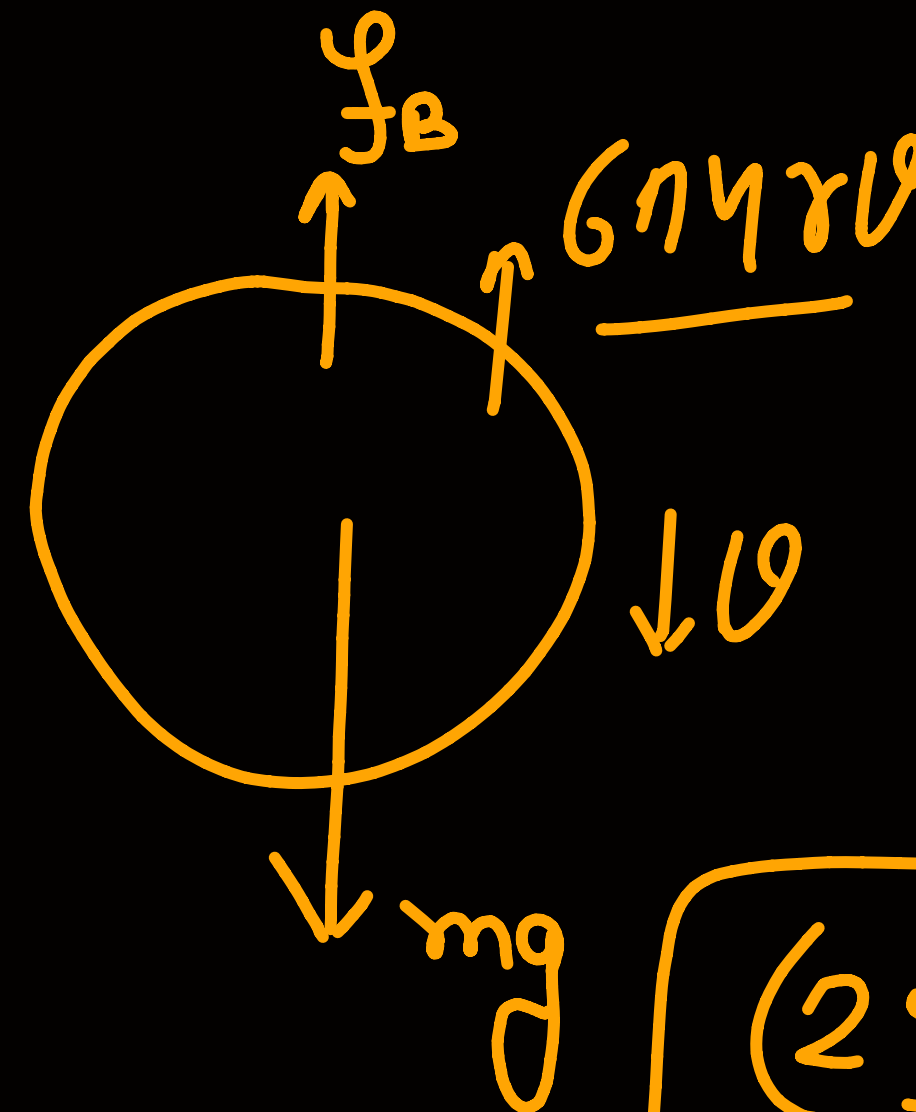
d_1 $\Rightarrow V_1 d_1 g - V_1 g = V_1 d_1 \frac{g}{2}$
 $d_1 - \frac{d_1}{2} = g$
 $\frac{d_1}{2} = g$
 $d_1 = 2g$



$$V_2 d_2 g - V_2 g = V_2 d_2 \frac{g}{4}$$

$$\frac{3d_2}{4} = g$$

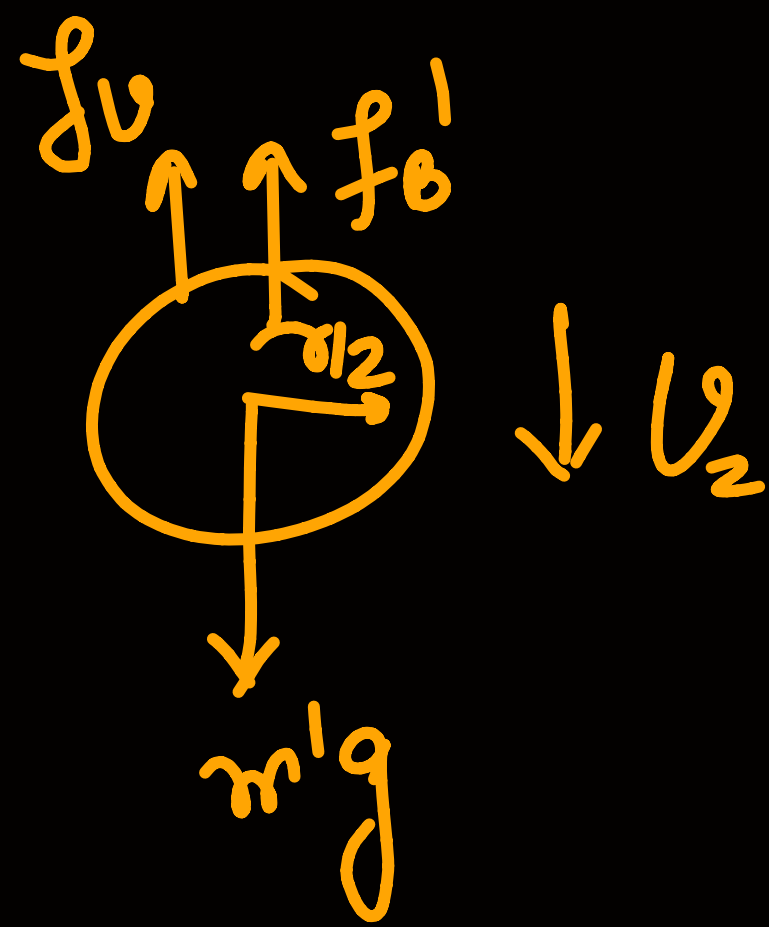
$$d_2 = \frac{4g}{3}$$



$$mg = f_B + 6\pi\eta r v$$

$$mg - f_B = \frac{mg}{2} = 6\pi\eta r v$$

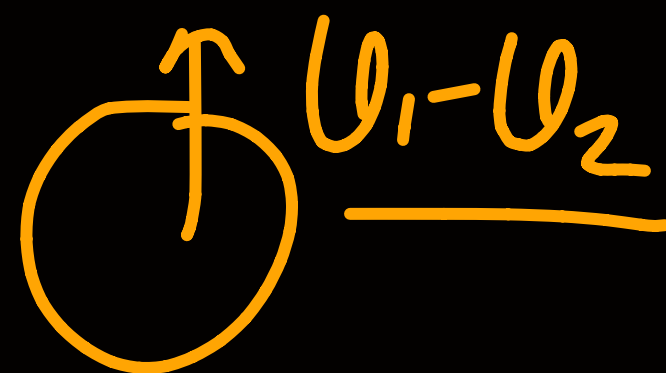
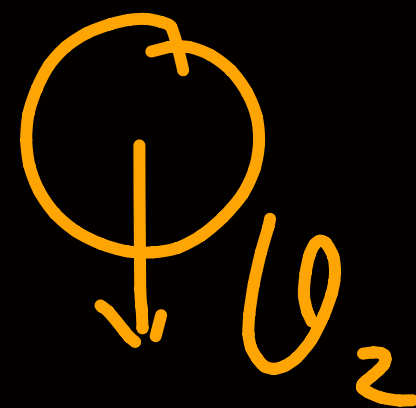
$$\frac{(2g)\frac{4}{3}\pi r^3 g}{2 \times 6\pi\eta r} = v_1$$



$$m'g = f_0' + 6\pi\eta r \frac{v_2}{2}$$

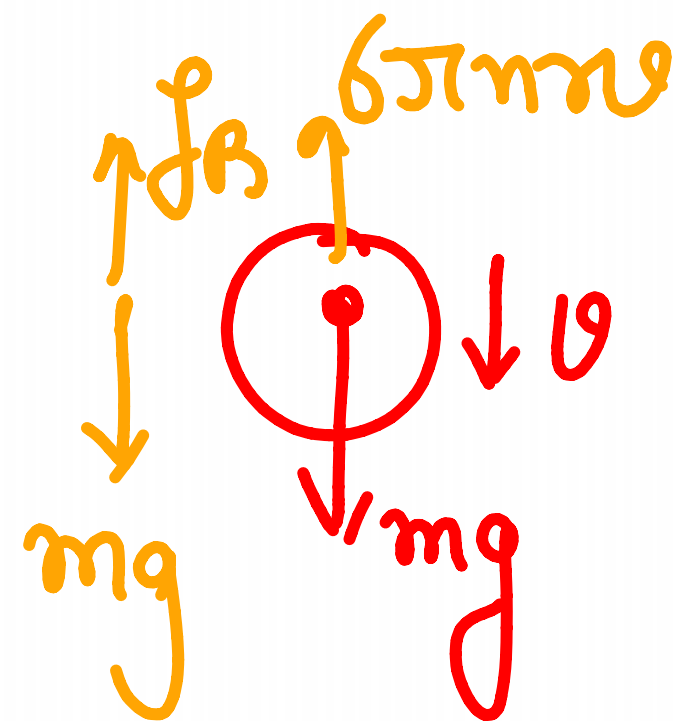
$$m'g - f_0' = \frac{m'g}{4} = 6\pi\eta r \frac{v_2}{2}$$

$$\frac{\left(\frac{49}{3}\right) \frac{4}{3} \pi \left(\frac{r}{2}\right)^3 g}{4 \times 6\pi\eta r \frac{r}{2}} = v_2$$



Q.9. A spherical ball of radius r and density d is dropped from rest in a viscous fluid having density ρ and coefficient of viscosity η .

- Calculate the power (P_1) of gravitational force acting on the ball at a time t after it is dropped.
- Calculate the rate of heat generation (P_2) due to rubbing of fluid molecules with the ball, at time t after it is dropped.
- How do P_1 and P_2 change if the radius of the ball were doubled?
- Find P_1 and P_2 when both become equal.



$$d \frac{4}{3} \pi r^3 g - \rho \frac{4}{3} \pi r^3 g - 6 \pi \eta r v = d \frac{4}{3} \pi r^3 \left(\frac{dv}{dt} \right)$$

$$\frac{(d - \rho) g}{2} - \frac{\eta v}{r^2} = d \frac{dv}{dt}$$

$$\boxed{\frac{dv}{dt} = \underbrace{\left(\frac{1 - \rho}{d} \right) g}_a - \underbrace{\frac{\eta v}{2 r^2 d}}_b}$$

$$v \frac{dv}{dt} = a - bv$$

$$\int_0^v \frac{dv}{a - bv} = \int_0^t dt$$



$$-\frac{1}{b} \ln(a-bv)_0^v = (t|_0^t$$

$$-\frac{1}{b} \ln \frac{a-bv}{a} = t$$

$$\frac{a-bv}{a} = e^{-bt}$$

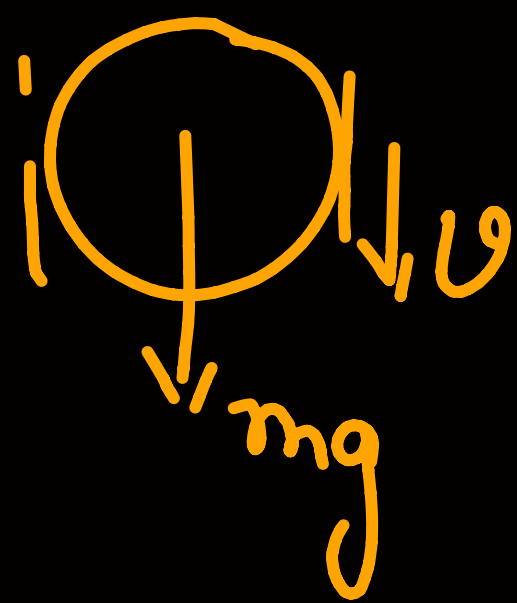
$$1 - \frac{b}{a}v = e^{-bt}$$

$$1 - e^{-bt} = \frac{b}{a}v$$

$$v = \frac{a}{b} \{1 - e^{-bt}\}$$

$$v = \frac{(1-\frac{f}{d})g2\tau^2d}{g\gamma} \left\{ 1 - e^{\frac{-g\gamma}{2\tau^2d}t} \right\}$$

$$v = \frac{2(d-f)g\tau^2}{g\gamma} \left(1 - e^{\frac{-g\gamma}{2\tau^2d}t} \right)$$



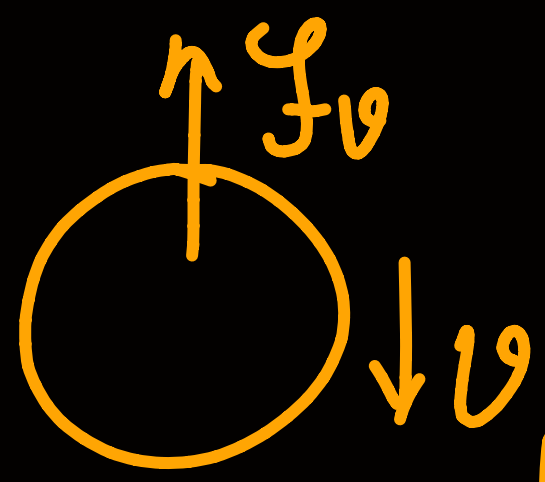
$$P_g = mgv \cos 0^\circ$$

$$= \frac{4}{3}\pi\tau^3d g \frac{2(d-f)g\tau^2}{2\gamma} \left(1 - e^{\frac{-g\gamma}{2\tau^2d}t} \right)$$

$$P \propto \tau^5$$

$$\tau \rightarrow (2\tau)$$

$$P \rightarrow 32P$$



$$P_{Fv} = F_0 v \cos \pi = -F_0 v$$

$$P_2 = \frac{dQ}{dt} = F_0 v = 6\pi \eta r v^2$$

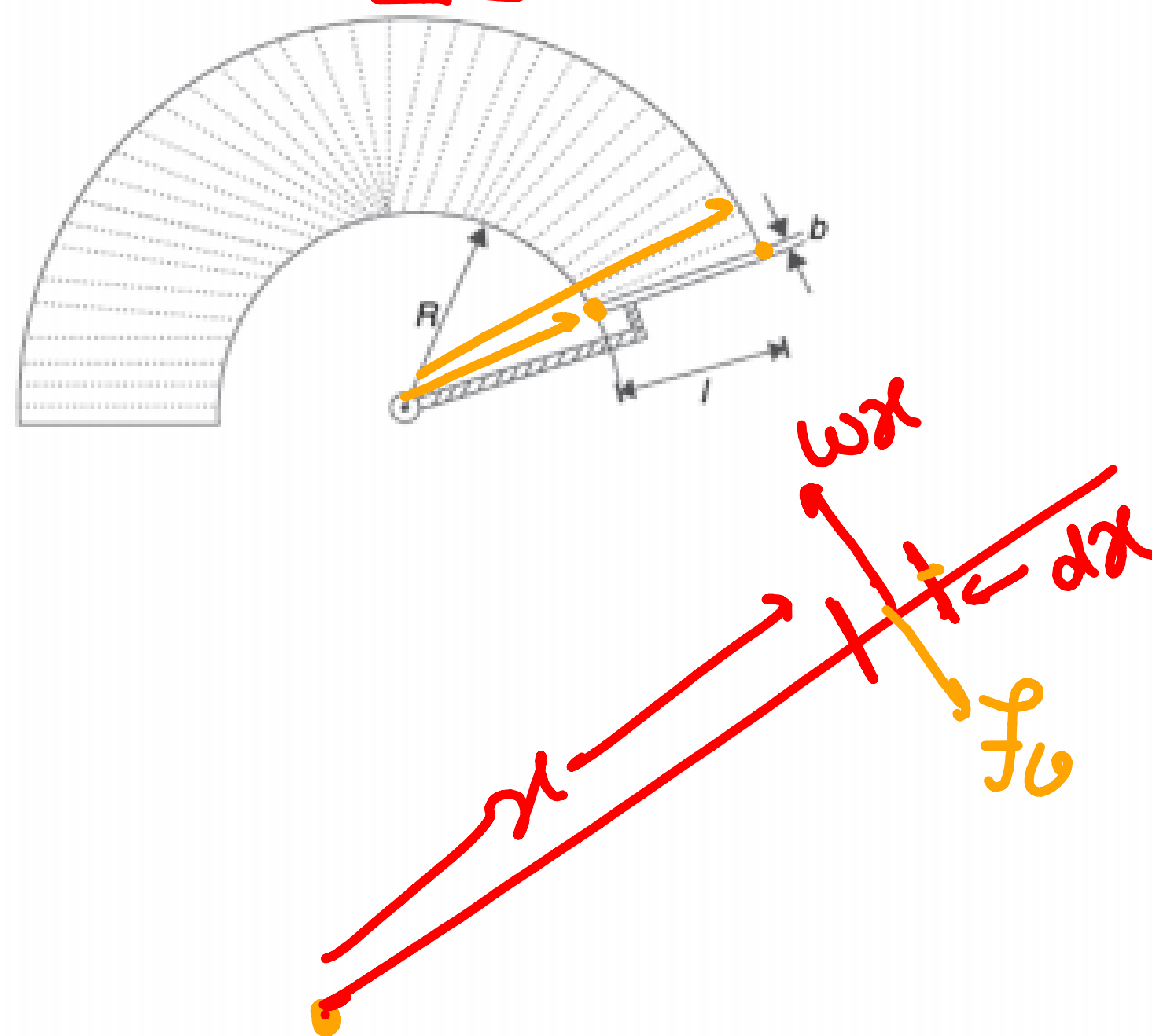
$$P_2 \propto r (r^4)$$

$$P \propto r^5$$

$$P \rightarrow 32P$$

$$mgv =$$

Q.11. A car windshield wiper blade sweeps the wet windshield rotating at a constant angular speed of ω . R is the radius of innermost arc swept by the blade. Length and width of the blade are l and b respectively. Coefficient of viscosity of water is η . Calculate the torque delivered by the motor to rotate the blade assuming that there is a uniform layer of water of thickness t on the glass surface.



$$f_v = (\eta b dx) \left(\frac{\omega x}{t} \right)$$

$$\tau = \int d\tau = \int_R^{R+l} \frac{\eta b \omega}{t} (x^2 dx)$$

$$= \frac{\eta b \omega}{3t} \left((R+l)^3 - R^3 \right) = \tau$$

Q.2. The elastic limit and ultimate strength for steel is $2.48 \times 10^8 \text{ Pa}$ and $4.89 \times 10^8 \text{ Pa}$ respectively. A steel wire of 10 m length and 2 mm cross sectional diameter is subjected to longitudinal tensile stress. Young's modulus of steel is $Y = 2 \times 10^{11} \text{ Pa}$

(a) Calculate the maximum elongation that can be produced in the wire without permanently deforming it. How much force is needed to produce this extension?

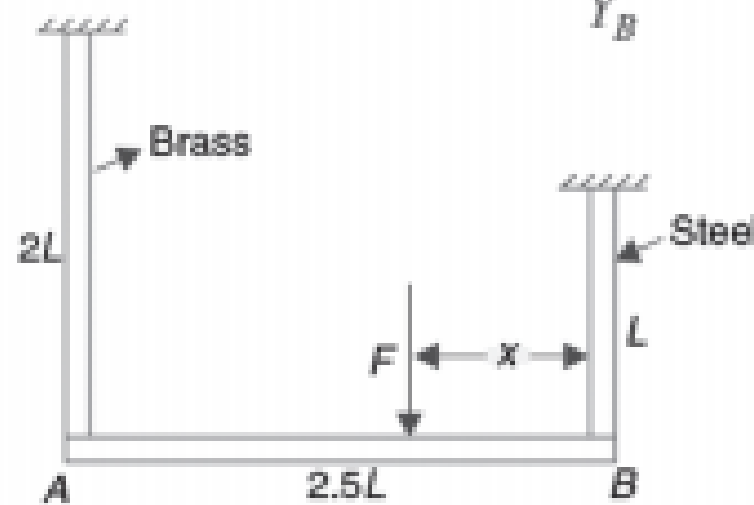
(b) Calculate the maximum stretching force that can be applied without breaking the wire.

$$2.48 \times 10^8 \frac{\text{N}}{\text{m}^2} = \frac{F}{\pi (10^{-3})^2}$$

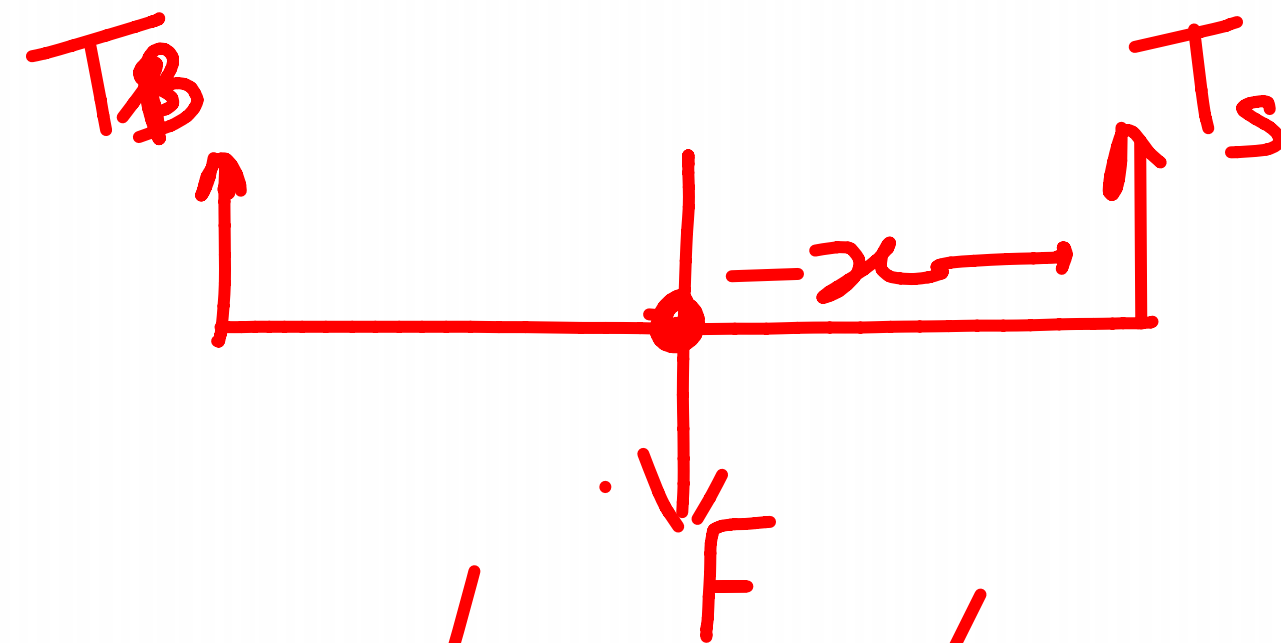
$$\frac{2.48 \times 10^8}{(0.2/10)} = 2 \times 10^{11}$$

$$4.89 \times 10^8 = \frac{F}{\pi (10^{-3})^2}$$

Q.6. A very stiff bar (AB) of negligible mass is suspended horizontally by two vertical rods as shown in figure. Length of the bar is $2.5L$. The steel rod has length L and cross sectional radius of r and the brass rod has length $2L$ and cross sectional radius of $2r$. A vertically downward force F is applied to the bar at a distance x from the steel rod and the bar remains horizontal. Find the value of x if it is given that ratio of Young's modulus of steel and brass is $\frac{Y_s}{Y_B} = 2$.



$$T_s + T_B = F$$



$$T_s x = T_B (2.5L - x)$$

$$\boxed{x = 1.25L}$$

$$y = \frac{T/A}{\Delta l/l} = \frac{T \cdot l}{\pi r^2 \Delta l}$$

$$\Delta l = \frac{T l}{\pi r^2 y}$$

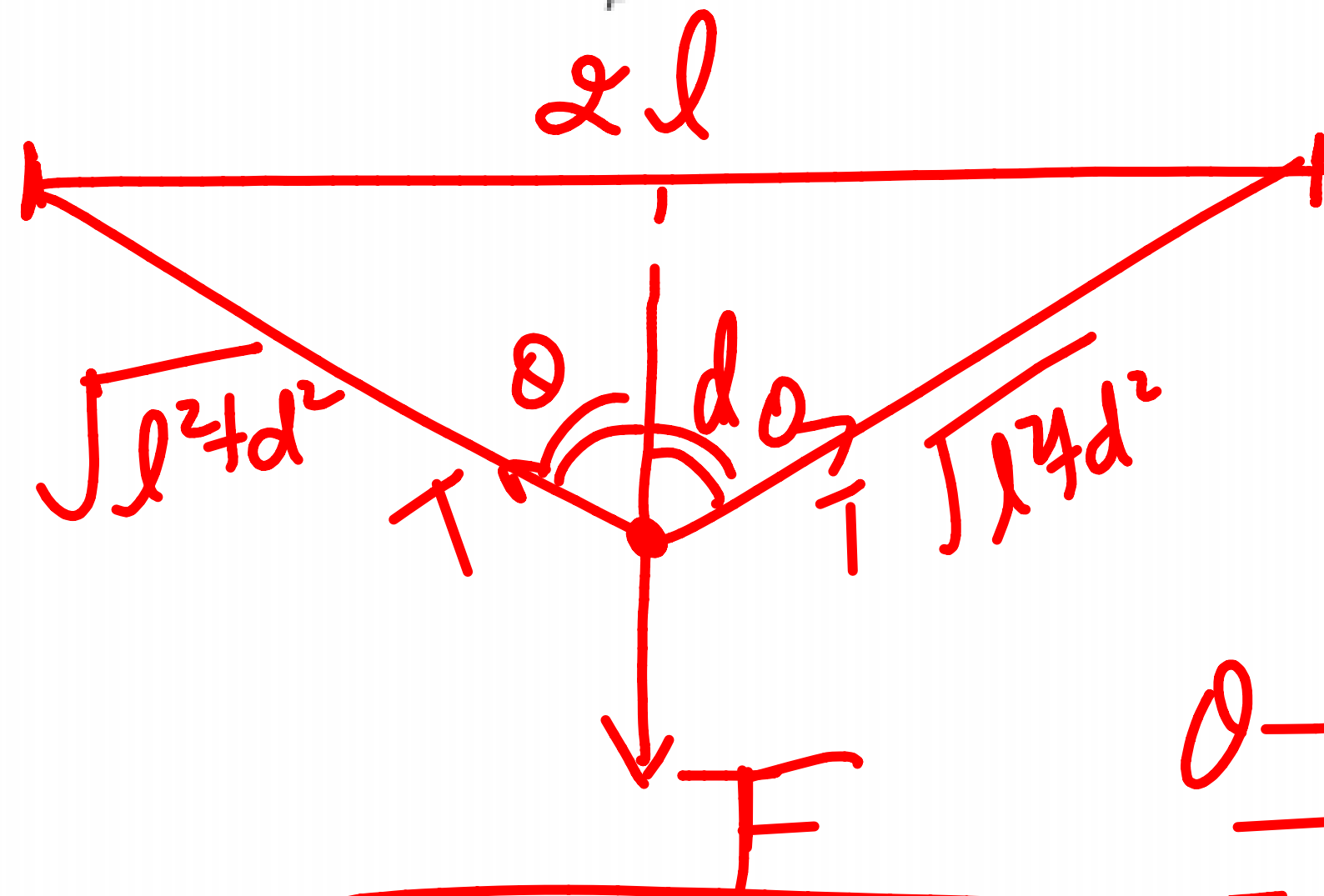
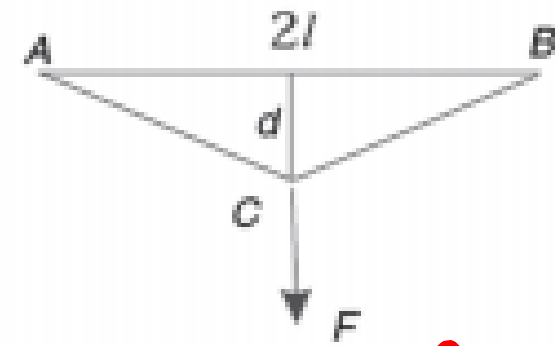
$$\Delta l_B = \frac{T_B \times 2L}{\pi (2r)^2 Y_B} = \Delta l_s = \frac{T_s \times L}{\pi r^2 \cdot 2 Y_B}$$

$$\frac{T_B}{2} = \frac{T_s}{2} \Rightarrow \boxed{T_B = T_s}$$

Q.10. A steel wire of radius r is stretched without tension along a straight line with its ends fixed at A and B (figure). The wire is pulled into the shape ACB . Assume that d is very small compared to length of the wire. Young's modulus of steel is Y .

(a) What is the tension (T) in the wire?

(b) Determine the pulling force F . Is F larger than T ?



$$F = 2T \cos \theta$$

$\theta \rightarrow 90^\circ$
 $F \ll 2T$
 $F \ll T$

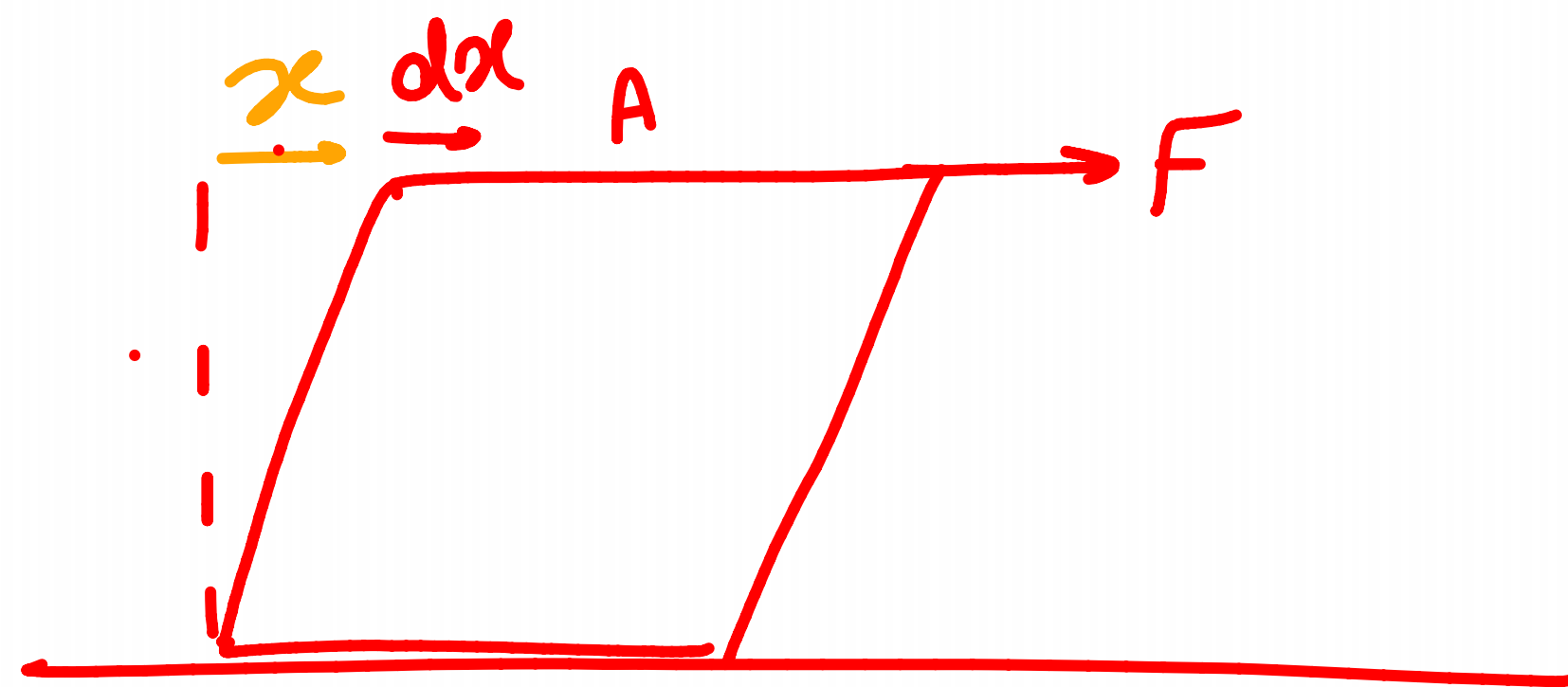
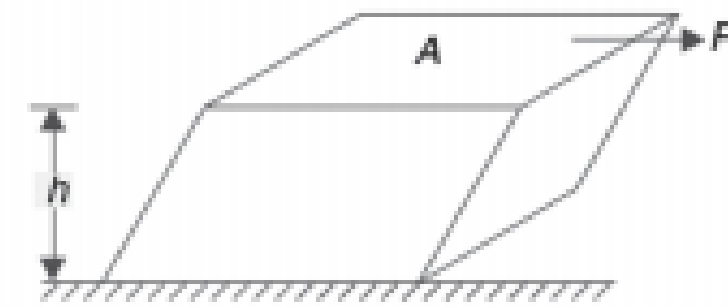
$$\begin{aligned} \Delta l &= 2\sqrt{l^2 + d^2} - 2l \\ &= 2l \left[\left(1 + \frac{d^2}{l^2}\right)^{\frac{1}{2}} - 1 \right] \\ &= 2l \left[1 + \frac{d^2}{2l^2} - 1 \right] \\ &= \frac{d^2}{l} = \Delta l \end{aligned}$$

$$Y = \frac{T/A}{\Delta l/l} = \frac{Tl}{A\Delta l} \Rightarrow T = \frac{YA\Delta l}{l}$$

$$T = \frac{Y\pi r^2 d^2}{(2l)l}$$

$$F = \frac{2T \cdot d}{\sqrt{l^2 + d^2}}$$

Q.12. A rectangular bar is fixed to a hard floor. Height of the bar is h and its area in contact with the floor is A . A shearing force distorts the bar as shown. Prove that the work done by the shearing force is $W = \left(\frac{\sigma^2}{2\eta} \right) \times \text{volume of the bar}$. Here σ is shear modulus of elasticity. Assume the deformation to be small.

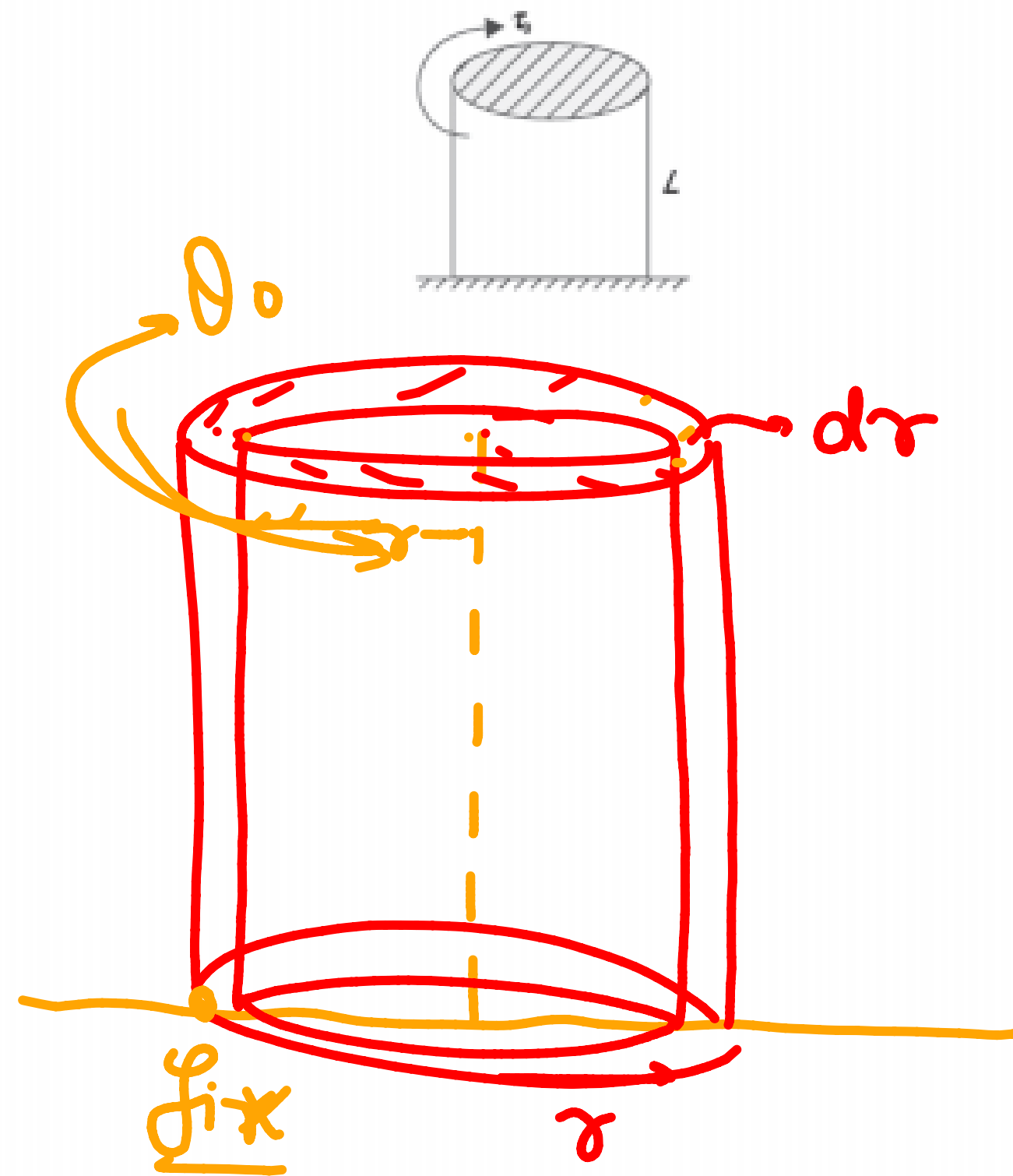


$$\sigma = \frac{Fh}{Ax} \Rightarrow F = \frac{\sigma A x}{h}$$

$$dw = \frac{\sigma Ax}{h} dx \cos 0$$

$$\begin{aligned} W &= \frac{\sigma A}{h} \int_0^x x dx \\ &= \frac{\sigma A}{h} \cdot \frac{x^2}{2} \\ &= \frac{\sigma (Ah)}{2} \left(\frac{x^2}{h^2} \right) \\ &= \frac{\sigma}{2} \left(\text{Strain} \right)^2 \text{Volume} \end{aligned}$$

Q.17. A metal cylinder of length L and radius R is fixed rigidly to ground with its axis vertical. A twisting torque τ_0 is applied along the circumference at the top of the cylinder. This causes an angular twist of θ_0 (rad) in the top surface. Calculate the shear modulus of elasticity (η) of the material of the cylinder.



$$\text{Strain} = \frac{\gamma \theta_0}{l}$$

$$\text{Stress} = \eta \times \text{strain} = \frac{\eta \gamma \theta_0}{l}$$

$$\frac{\text{torque}}{\text{Area}} = \frac{\eta \gamma \theta_0}{l} \gamma = \frac{\eta \gamma^2 \theta_0}{l}$$

$$Z = \int dZ = \int \frac{\eta \gamma^2 \theta_0}{l} (2\pi r dr)$$

$$\tau_0 = \frac{\eta \theta_0 2\pi}{l} \int_0^R r^3 dr \Rightarrow \eta = \frac{2 \tau_0 l}{\pi \theta_0 R^4}$$

$$\tau_0 = \frac{\pi \eta \theta_0 R^4}{2l}$$

Q.16. Atmospheric pressure is P_0 and density of water at the sea level is ρ_0 . If the bulk modulus of water is B , calculate the pressure deep inside the sea at a depth h below the surface.

$$m = \rho v$$

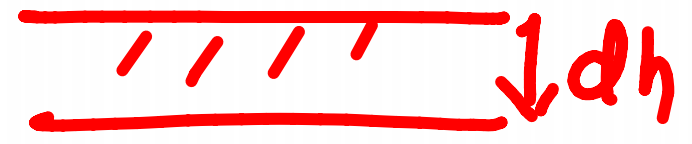
$$0 = \rho dv + v d\rho$$

$$\frac{d\rho}{\rho} = -\frac{dv}{v}$$

$$B = -\frac{dP}{dv/v}$$

$$B = +\frac{dP}{d\rho/\rho}$$

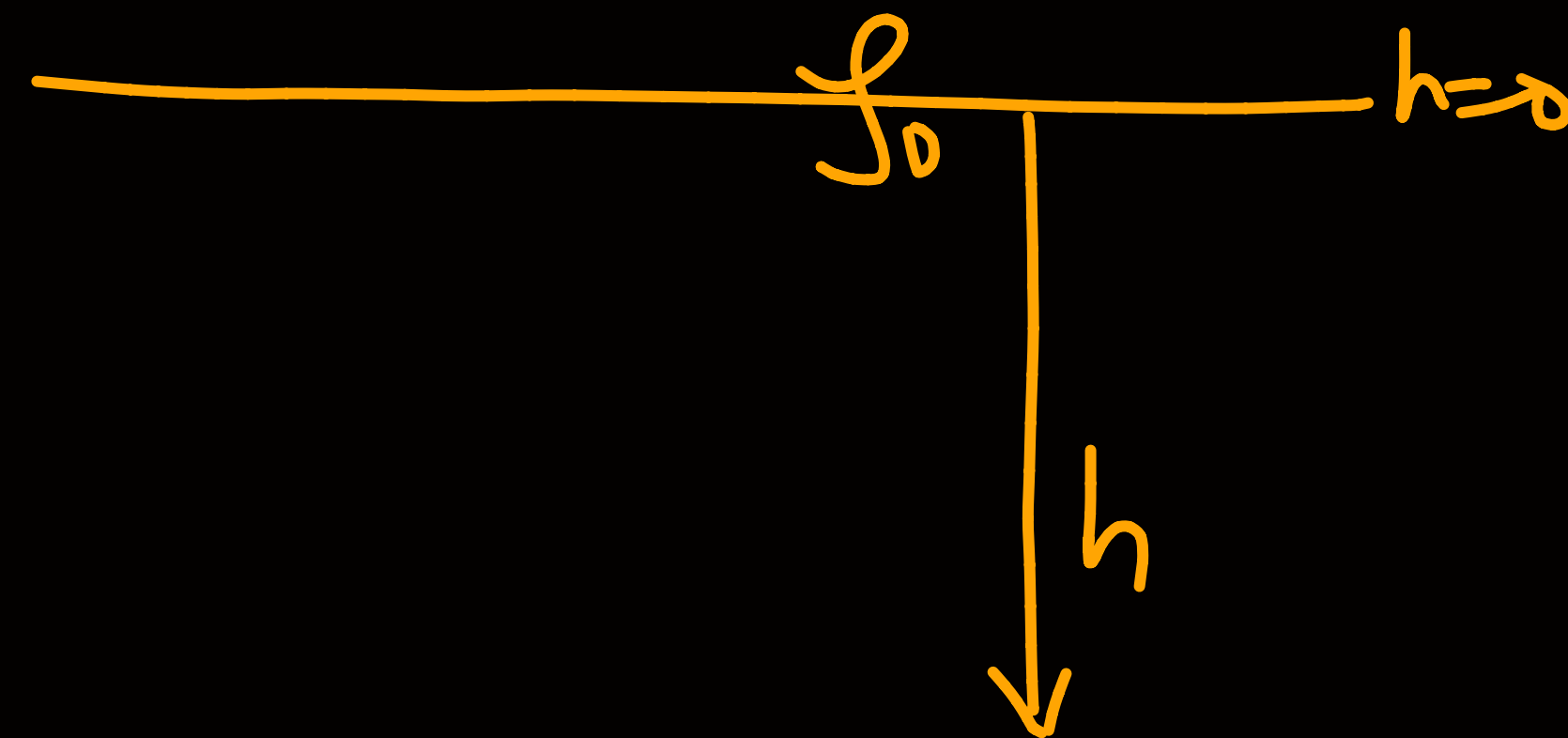
$$dP = B \frac{d\rho}{\rho}$$



$$dP = \rho g dh$$

$$\int_{\rho_0}^{\rho} \frac{d\rho}{\rho^2} = \frac{g}{B} \int_0^h dh$$

$$\rho g dh = B \frac{d\rho}{\rho}$$



$$\left(-\frac{1}{\rho} \right)_{\rho_0}^{\rho} = \frac{g}{B} (h)_0^h$$

$h, \rho \uparrow$

$$\frac{1}{\rho_0} - \frac{1}{\rho} = \frac{gh}{B}$$

$$\frac{1}{\rho} = \frac{1}{\rho_0} - \frac{gh}{B}$$

$$dp = B \frac{dy}{y} =$$

$$y = \frac{f_0 B}{B - f_0 g h}$$

$$\frac{1}{y} = \frac{1}{f_0} - \frac{g h}{B}$$

$$+ \frac{1}{y^2} dy = 0 + \frac{g}{B} dh$$

$$\boxed{\frac{dy}{y} = \frac{y g}{B} dh}$$

$$\int_{P_0}^P dp = \int_0^h \frac{f_0 B g}{B - f_0 g h} dh$$

$$(P)_P = \frac{f_0 B g}{-f_0 g} \ln(B - f_0 g h)_0^h$$

$$P - P_0 = -B \ln \frac{B - f_0 g h}{B}$$

$$P = P_0 + B \ln \frac{B}{B - f_0 g h}$$